

Research Project Summary

A Comprehensive Evaluation of Machine Learning Models for Skincare Ingredient Classification with Class Sparsity Reduction

78.6%	77.2%	+21.4 pts
reported classification accuracy	reported precision / recall	reported accuracy gain after CSR

Research question

How can supervised machine-learning models classify cosmetic-ingredient functions when the underlying labels are sparse, multi-label, and affected by non-standard INCI and IUPAC naming?

Data and problem setting

The study uses the European Commission CosIng ingredient data and related regulatory annexes. The task maps ingredient names and descriptions to functional labels. The central challenge is class sparsity: many relevant functions have too few examples for a conventional classifier to learn reliable decision boundaries.

Method

The project introduces Class Sparsity Reduction (CSR), a domain-informed grouping strategy that consolidates sparse ingredient-function labels into broader functional categories. The reproducible notebook evaluates Logistic Regression, Decision Trees, Random Forest, and XGBoost using TF-IDF features from ingredient names and descriptions. The accompanying training service contains the production-oriented multi-label inference workflow and saved model artifacts.

Reported findings

The project materials report 78.6% classification accuracy, 77.2% precision/recall, and a 21.4-point accuracy improvement after CSR. These figures should be interpreted alongside the included notebook, dataset documentation, and paper, which provide the experimental context and limitations.

Research artifacts included in this repository

paper/ contains the accompanying research paper and this summary. **notebooks/** contains the executed notebook and static HTML export. **data/** contains the CosIng source materials used by the notebook. **src/** contains the training and inference-support code, requirements, and exported model artifacts. The deployed product implementation lives in the separate CUTleS-IQ repository.

Reproducibility and scope

Run the notebook from the repository root after installing the documented Python dependencies. The repository is intended for educational and research review. It is not a clinical diagnostic system and does not provide medical advice.

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